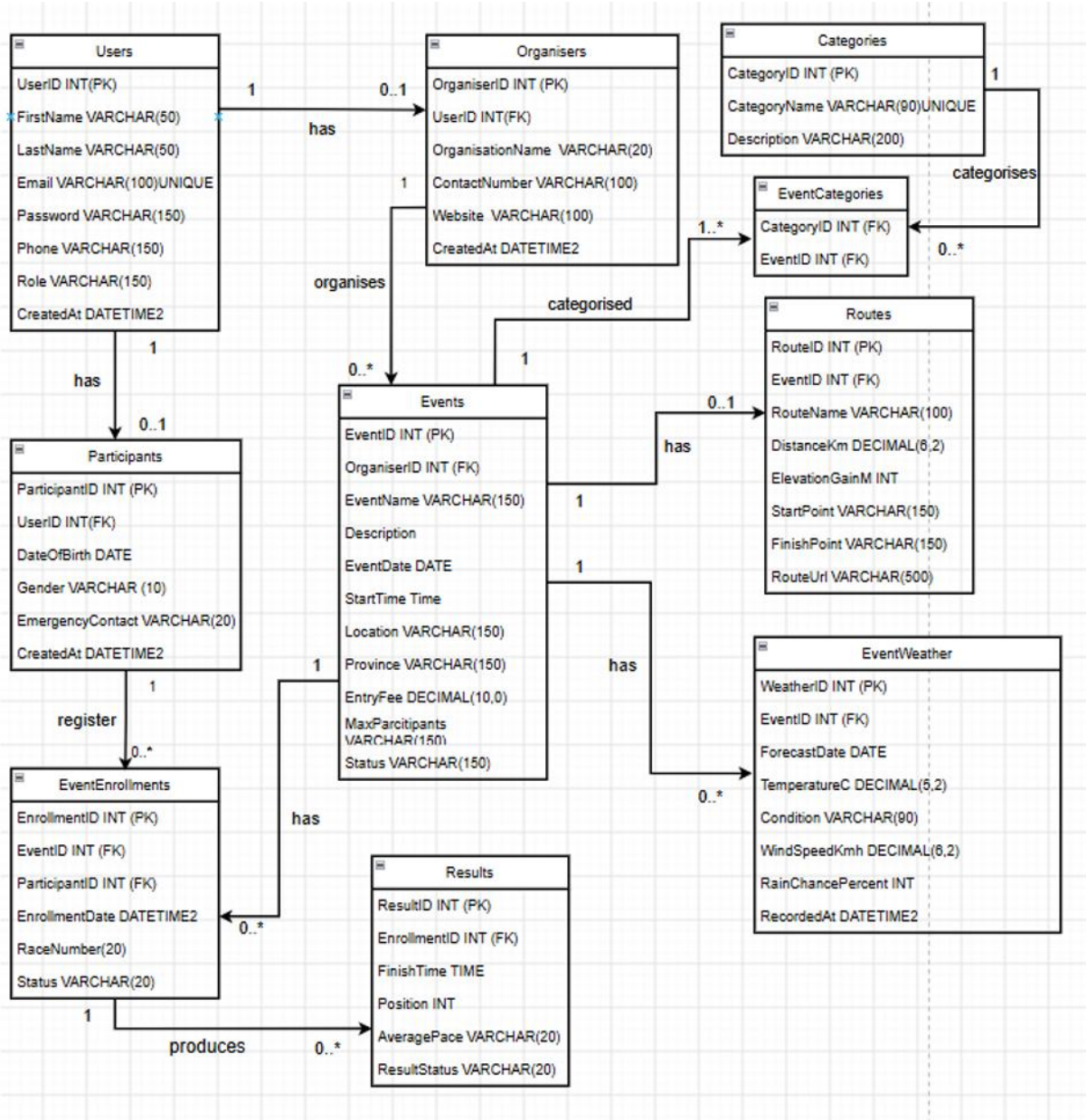
Section A

ERD:



Business Rules:

-A User can be associated with a maximum of one Organiser profile, and each Organiser profile must be connected to one User.

-A User can have at most one Participant profile, with every Participant profile belonging to a single User.

-An Organiser can be responsible for several Events, but each Event can only be assigned to one Organiser.

-Participants can take part in multiple Events, and an Event can accommodate multiple Participants.

-A Participant is permitted to enrol in each Event only once.

-Every Event must be linked to one or more Categories, while a Category can be linked to numerous Events.

-An Event must not contain duplicate Category assignments.

-An Event can exist without a Route or with one Route, but a Route cannot be associated with more than one Event.

-Each Event can have multiple Weather records, although every Weather record must correspond to one Event.

-An Event can contain several EventEnrollments, with each enrollment linked to one specific Event.

-Every EventEnrollment must be associated with one Participant.

-The number of Participants enrolled in an Event must remain within the Event's MaxParticipants limit.

-A Participant's RaceNumber cannot be duplicated within the same Event.

-An EventEnrollment can have no Result or one Result, while each Result must belong to one EventEnrollment.

-Foreign-key fields must always contain values that exist in the referenced table, ensuring that the relationships between records remain valid.

Section B

1. Event Organiser API Endpoints

HTTP Method	Route	Purpose / Description	Required Role	Request Data	Possible Response
POST	/api/auth/register	Registers a new account for an event organiser.	Event Organiser	{ name, email, password, role }	201 Created – account registered. 400 Bad Request – incorrect information. 409 Conflict – email is already registered.
POST	/api/auth/login	Checks the organiser's login details and issues an authentication token.	Event Organiser	{ email, password }	200 OK – authentication successful and token supplied. 401 Unauthorized – login details are invalid.
GET	/api/users/me	Displays the profile details belonging to the currently signed-in organiser.	Event Organiser	None	200 OK – organiser information returned. 401 Unauthorized – user must be authenticated.
PUT	/api/users/me	Changes the personal information of the authenticated organiser.	Event Organiser	{ name, email, phone, location }	200 OK – profile changes saved. 400 Bad Request – supplied details are invalid.
GET	/api/categories	Lists the categories that an organiser can select for an event.	Event Organiser	None	200 OK – available categories returned.
POST	/api/categories	Adds a new category to the event category list.	Event Organiser	{ name, description }	201 Created – category added. 400 Bad Request – category data is invalid.

PUT	/api/categories/{id}	Edits the details of an existing category.	Event Organiser	{ name, description }	200 OK – category details changed. 404 Not Found – requested category was not found.
DELETE	/api/categories/{id}	Removes an existing category from the system.	Event Organiser	None	204 No Content – category removed. 404 Not Found – category was not found.
GET	/api/events	Shows the events belonging to the authenticated organiser.	Event Organiser	None	200 OK – organiser's event list returned.
GET	/api/events/{id}	Provides complete information for one selected event.	Event Organiser	None	200 OK – event information returned. 404 Not Found – selected event was not found.
POST	/api/events	Adds and publishes a new running, walking or cycling event.	Event Organiser	{ name, description, categoryId, date, startTime, location, distance, routeInfo }	201 Created – event successfully added. 400 Bad Request – event details are invalid.
PUT	/api/events/{id}	Changes the details of an event controlled by the organiser.	Event Organiser	{ name, description, categoryId, date, startTime, location, distance, routeInfo }	200 OK – event updated. 404 Not Found – event unavailable. 403 Forbidden – organiser is not responsible for the event.
DELETE	/api/events/{id}	Deletes an event that is managed by the organiser.	Event Organiser	None	204 No Content – event removed. 404 Not Found – event could not be found.
GET	/api/events/{id}/enrolments	Lists all participants who have signed up for the selected event.	Event Organiser	None	200 OK – enrolled participants returned. 404 Not Found – event could not be found.
POST	/api/events/{id}/results	Adds a participant's race result once the event has finished.	Event Organiser	{ userId, finishTime, position, status }	201 Created – result saved. 400 Bad Request – result information is invalid. 404 Not Found – participant or event unavailable.
GET	/api/events/{id}/results	Displays the recorded results for everyone in a selected event.	Event Organiser	None	200 OK – event results returned. 404 Not Found – event was not found.
PUT	/api/results/{id}	Modifies an existing participant result when a correction is required.	Event Organiser	{ finishTime, position, status }	200 OK – result changed. 404 Not Found – result could not be found.

2. Participant API Endpoints

HTTP Method	Route	Purpose / Description	Required Role	Request Data	Possible Response
POST	/api/auth/register	Creates a Participant account that can be used to join events and monitor race performance.	Participant	{ name, email, password, role }	201 Created – account registered. 400 Bad Request – registration information is invalid. 409 Conflict – email is already registered.
POST	/api/auth/login	Verifies participant credentials and returns an authentication token.	Participant	{ email, password }	200 OK – login completed and token returned. 401 Unauthorized – incorrect credentials.
GET	/api/users/me	Returns the profile details of the participant currently signed in.	Participant	None	200 OK – participant profile returned. 401 Unauthorized – authentication is required.
PUT	/api/users/me	Allows the authenticated participant to edit their profile details.	Participant	{ name, email, phone, location }	200 OK – profile successfully changed. 400 Bad Request – invalid information supplied.
GET	/api/categories	Shows the event categories available to participants, including running, walking and cycling.	Participant	None	200 OK – category list returned.
GET	/api/events	Provides participants with a list of upcoming events they can view.	Participant	None	200 OK – upcoming events displayed.

GET	/api/events/{id}	Returns the information and details for a particular event.	Participant	None	200 OK – selected event details returned. 404 Not Found – event was not found.
POST	/api/events/{id}/enrolments	Signs the logged-in participant up for the chosen event.	Participant	{ emergencyContact, selectedDistance }	201 Created – registration completed. 404 Not Found – event does not exist. 409 Conflict – participant is already registered.
GET	/api/enrolments	Shows all events where the current participant has an active enrolment.	Participant	None	200 OK – participant's registered events returned.
DELETE	/api/enrolments/{id}	Cancels a participant's registration for an event.	Participant	None	200 OK – enrolment information returned. 404 Not Found – enrolment was not found.
GET	/api/users/me/results	Displays the participant's recorded race history and performance information.	Participant	None	200 OK – personal results returned. 401 Unauthorized – authentication is required.
GET	/api/events/{id}/weather	Returns weather details for the event to assist participants with race-day preparation.	Participant	None	200 OK – event weather information returned.